MARGO BONAL

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Education

Bachelor of Science – Computer Science Pennsylvania Western University, California PA

(GPA: 3.72, Expected Graduation Date: May 2026) *Magna Cum Laude Latin Honors*

Course Work

CMSC 3040 – Data Structures
CMSC 3140 – Analysis of Algorithms
CMSC 2040 – Object-Oriented Programming
CMSC 3180 – Computer Networking
CMSC 4000 – Operating Systems
CMSC 4080 – Structures of Program Language

Experience

Plexus Parc., Remote Internship

Software Developer Intern (Oct. 2025- Present)

- Collaborated with the Director of Technology and a cross-functional team to build and maintain Data Modeling software applications using modern frameworks.
- Developed front/back-end features with secure API integration, contributing to scalable systems.
- **Technologies:** Python (Flask), Java, HTML/CSS/JavaScript, MSSQL, Azure DevOps, EVBS Encryption, AWS, GitHub, SSH, API integration, MCP server, Markdown, LaTeX.

Encompass Remote INC., Hybrid Internship

Full Stack Developer Intern (May 2025- August 2025)

- Collaborated with the Director of Technology and a cross-functional team to build and maintain HIPAA-compliant medical software applications using modern frameworks.
- Developed front- and back-end features with secure API integration, contributing to scalable, reliable healthcare systems.
- **Technologies:** Python (Flask), Java, HTML/CSS/JavaScript, React, Node.js, TypeScript, Spring Boot, Maven, Docker, MSSQL, Azure DevOps, EVBS Encryption, AWS, GitHub, SSH, TensorFlow/PyTorch, API integration, MCP server, Markdown, LaTeX.

Encompass Remote INC. & Laura Ohlund (Carnegie Mellon University), Remote Internship

Software Engineer Intern (July 2024)

- Built an internal file-sharing platform supporting secure transmission and storage of medical-grade documents.
- Implemented encryption practices aligned with HIPAA, cybersecurity, and Hi-Trust standards.
- Developed front- and back-end integrations, APIs, libraries, and user interface components.
- Collaborated with a cross-functional intern team and documented system architecture.
- Presented the final product to IT leaders and strategic partners for roadmap integration.

Skills

- ***Languages:*** C++, C, Python, HTML, CSS, JavaScript, COBOL, Java, API, Prolog, Assembly
- ***Environments:*** Linux OS, Windows OS, Virtual Box, Docker, WSL
- ***Development Tools:*** Vs Code, Visual Studio, Dev C++, MinGW, MSYS, MySQL, MSSQL, Git/GitHub, Azure DevOps, Hibernate, Flask, Vite, Vector Database, Node.js, React, Maven, EVBS, NI Multisim, KiCad
- ***Technical:*** Soldering, Wire Wrapping

Awards/Societies

- CS Advisory Board Student Representative (2024-Present)
- Vulcan Merit Scholarship (2022 - Present), Dean's List/ Highest Honors (2022 - Present
- United States Dressage Federation Member, 65 hours of volunteer work through 4-H

Project Highlight

CareSync - CardioLink (May 2025- August 2025)

Encompass Remote Internship Project

- A specialized branch of the CareSync application, designed to manage cardiac implantable and wearable devices, as well as associated patients and device data.
- EVBS, Envelope Verification Based Security – encrypted login
- Docker Containerized MSSQL Database

Vulcan Activity Tracker (Aug 2025- May 2026)

Pennsylvania Western University Senior Project

- University Athletic Activity Tracker
- Student credentials required, email verification
- SHA-256 & Argon2– encrypted login
- Dynamic dashboard, highlights user features
- Interactive map & weather APIs feature for route planning
- Social features allow viewing/interacting with friends
- Leaderboard, users can compete with activity metrics
- Docker Containerized MSSQL Database, HTML, CSS, JS, Python engine

Bank Management System (July 2024-August 2024)

Individual Project, Language: Java

- Simulate bank management system with oriented approach, 13 file/class
- Login process, ATM graphics, mini statement, withdraw/deposits
- Implement MySQL database to store user login credentials
- Implement Swing and APIs

Mr. Cowboy (March 2024- May 2024)

Group Project, Language: Python

- 2D Platform Game, Arena Battle
- Created original concept art (Sprite Frames/environmental)
- Multi-filed object-oriented design, implemented classes for modular actions
- Game physics/ frame rate/ multiple game states

Technical:

Intel 8085 Single Board Computer (Jan 2024- May 2024)

Project led by Dr. Anthony S. Pyzdrowski

Skills Acquired:

- Reading/ Interpreting Electrical Component Datasheets
- Create Schematics, Wiring Diagram, PCB Layout
- Soldering, Wire Wrapping
- Programming/ Debugging UARTs